

SENTINEL

Technical Workflow & Integration Diagram Specification

Gujarat CCTV Integration Hackathon 2026 | Category 1 (Academic / Research / Startup)

Team Name: KodeMatrix

Document: Workflow & Integration Diagram Specification

Project: Sentinel: Statewide CCTV Integration & ANPR Surveillance Engine

Repository: [DevamShah1211/Sentinel-Hackathone](#)

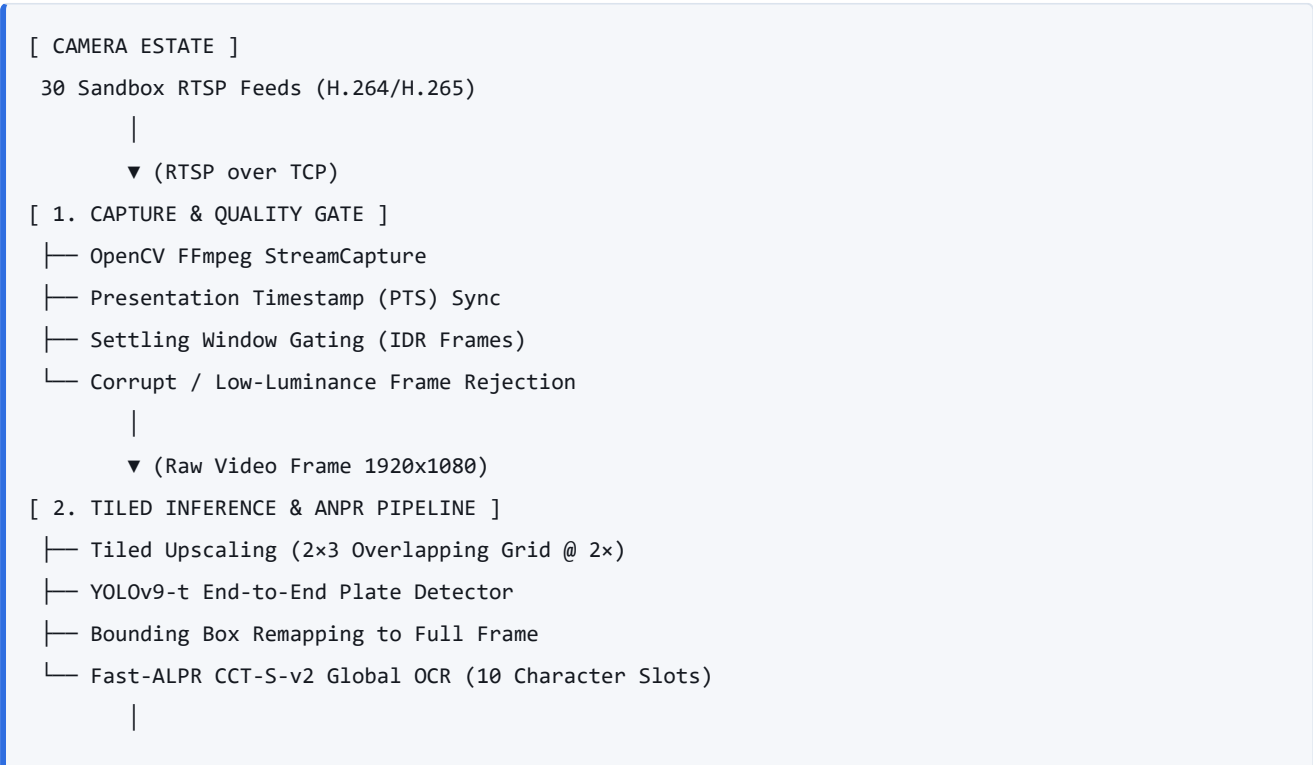
1. Executive Summary

Sentinel is designed as a modular, edge-first statewide CCTV integration platform that ingests camera streams from heterogeneous departmental network feeds, performs continuous local license plate recognition (ANPR), correlates detections against real-time enforcement watchlists, dispatches low-latency WebSocket notifications, and reconstructs vehicle travel corridors on GIS maps with comprehensive audit logging.

This document details the operational workflow, end-to-end data pipelines, microservice integration points, database schemas, and edge-to-cloud scalability topology.

2. End-to-End Operational Workflow Diagram

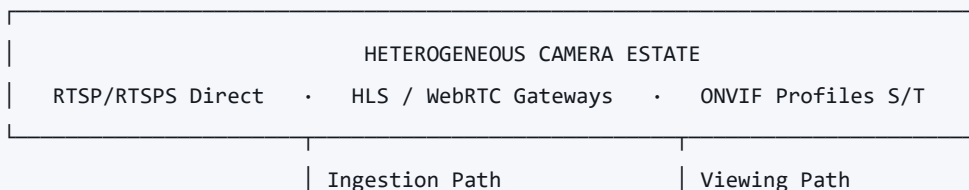
The sequence below illustrates how a live RTSP feed is ingested, processed, matched against enforcement watchlists, and surfaced to operators:

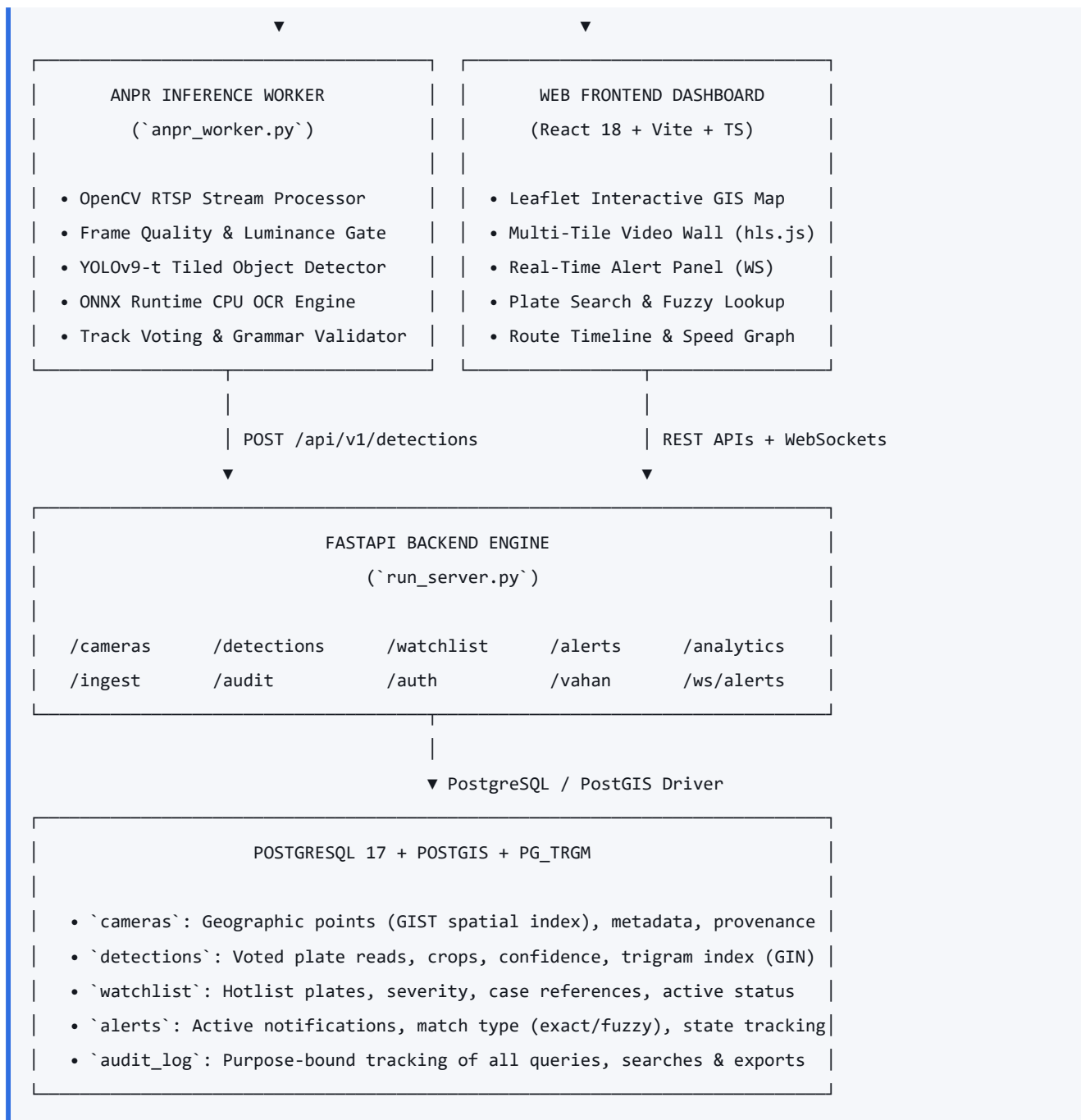


- ▼ (Per-Frame Plate Candidates)
- [3. NOISE REJECTION & TRACK VOTING]
 - └ Camera Overlay Text Rejection (Timestamps, PTZ Labels, Billboards)
 - └ IoU Spatial-Temporal Track Association (20-60 Frames per Vehicle)
 - └ Character-Level Confidence-Weighted Voting (Right-Aligned Serial)
 - └ Indian Plate Grammar Correction (O↔0, I↔1, S↔5, B↔8, Z↔2, G↔6)
 - |
 - ▼ (Validated Voted Plate Detection)
- [4. FASTAPI INGEST & DB COMMIT]
 - └ REST POST /api/v1/detections (Camera ID, Plate Text, Crop URI, PTS)
 - └ PostgreSQL 17 + PostGIS + pg_trgm Transaction Commit
 - └ Evidence Crop Saved to Local Storage (`/static/crops/`)
 - |
 - ▼ (Synchronous Pre-Commit Trigger)
- [5. WATCHLIST CORRELATION & ALERTING]
 - └ Exact String Match Check (Score: 1.0)
 - └ Trigram Fuzzy Match Check (`pg\_trgm` Similarity > 0.7)
 - └ Alert Row Created (`status = new`, Severity: Critical/High/Med)
 - └ WebSocket Manager Fan-out (`ws://.../ws/alerts`)
 - |
 - ▼ (Live Alert Notification)
- [6. OPERATOR UI & WORKFLOW ACTION]
 - └ React + Leaflet Live Map Toast & Audio Alert
 - └ Video Wall Tile Highlight (HLS Grid + Hero WebRTC)
 - └ Operator Acknowledgment (`status = ack`, Stated Purpose)
 - └ VAHAN Registry Enrichment (Owner, Make, Model, Fuel Type)
 - |
 - ▼ (Investigation & Reporting)
- [7. ROUTE RECONSTRUCTION & AUDIT]
 - └ Chronological Sighting Query & GIS Point Ordering
 - └ Great-Circle Distance, Time & Implied Speed Calculation
 - └ Speed Violation Flagging (>150 km/h Impossible Transition Flag)
 - └ Purpose-Bound Audit Logging (`audit\_log` Table)
 - └ Excel (.xlsx) & PDF Output Evidence Report Export

3. Microservice & System Integration Topology

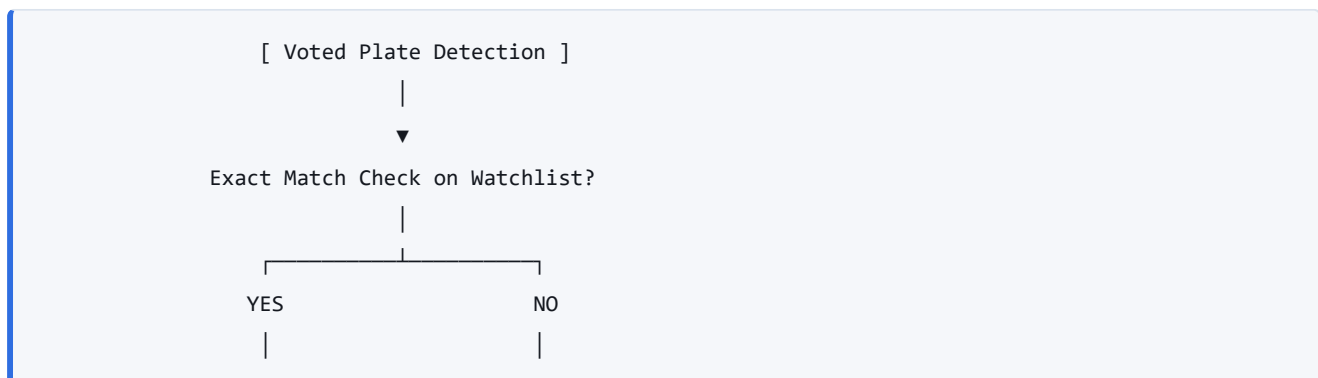
Sentinel separates video ingestion and compute-heavy inference from the API web backend and frontend UI layer:

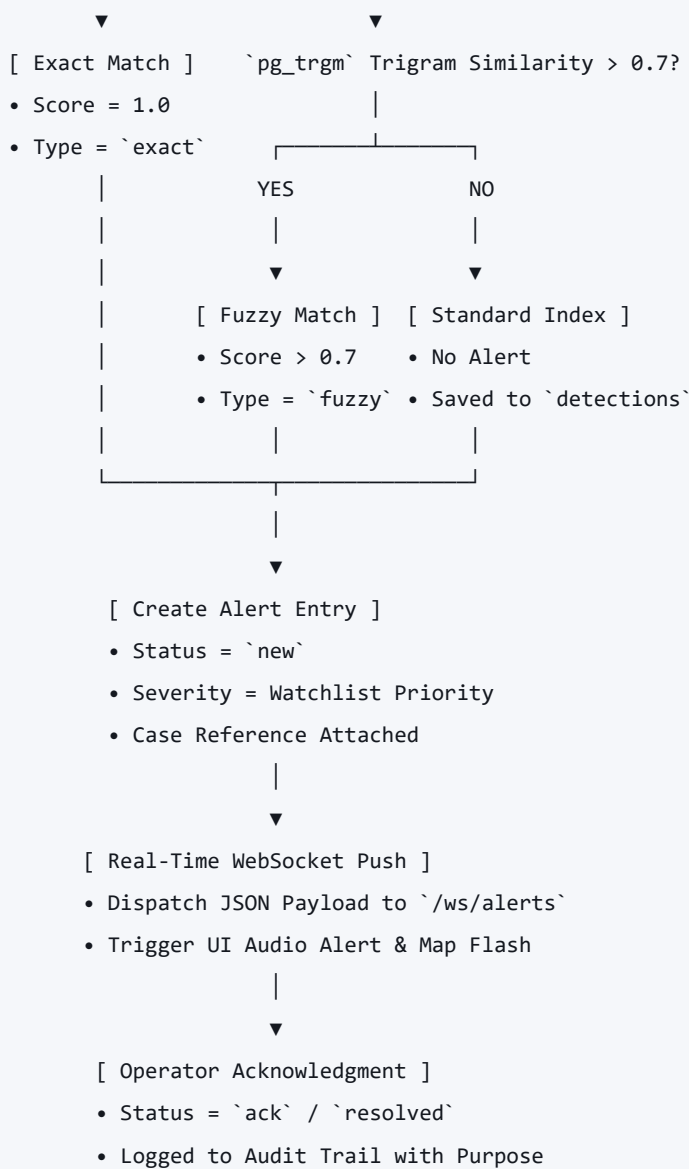




4. Watchlist Correlation & Alert Workflow

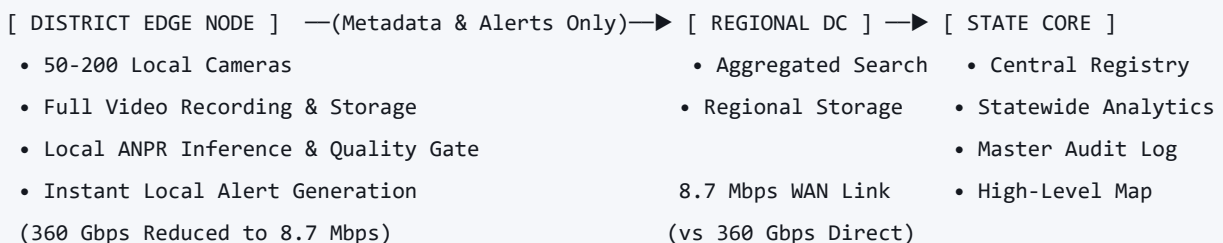
The diagram below details the exact decision tree executed whenever a number plate is indexed:





5. Statewide Edge-First Deployment Topology (Scale to 80,000 Cameras)

To prevent centralized network backhaul saturation (192 Gbps raw video bottleneck), Sentinel utilizes an **edge-first architecture**:



6. Integration Contract Interfaces

6.1 VAHAN Vehicle Registry Adapter Interface

- **Endpoint:** GET /api/v1/vahan/vehicle/{plate_number}
- **Mock / Integration Adapter:** Contract-first architecture returning registration date, owner name, vehicle class, fuel type, engine number, and insurance validity.

6.2 Audit Trail Verification Interface

- **Endpoint:** GET /api/v1/analytics/audit
- **Enforcement:** Logs user ID, role, client IP, action type (SEARCH , ROUTE_RECONSTRUCT , EXPORT), timestamp, and explicit **stated legal purpose**.

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